Sudo Thermal Camera: Heat-Adjustable Generative Video from a Single RGB Image

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Abstract

We present Sudo Thermal Camera, a training-free system that takes a single RGB image and a user-specified heat-source map as input and generates a physically plausible video showing what the scene would look like if that heat were applied: water boils, ice melts, wood chars, steel glows. Our framework combines four components, none of which are learned: (i) text-prompted material segmentation with SAM3, (ii) a 2D pixel-space thermal simulator implementing Fourier heat diffusion with phase change and reaction progress, (iii) heat-aware Cross-Attention Guidance (CAG) that steers a frozen image-to-video diffusion transformer localized by the simulator's evolving thermal mask, and (iv) pixel-composite preservation that locks non-heated pixels bit-equal to the input image. Unlike conventional adapter-based methodologies that require extensive paired thermal datasets and often suffer from run-to-run variance or residual-leak artifacts, our proposed inference-time attention steering achieves robust, reproducible control with zero trainable parameters. We extensively validate our framework across diverse thermodynamic scenarios, including liquid vaporization, wood combustion, metal incandescence, wax ignition, and paper degradation—demonstrating highly realistic, scene-appropriate generative dynamics. Furthermore, by analyzing complex out-of-distribution phase-change scenarios, we provide a critical investigation into the interaction between physics-informed constraints and the pre-existing visual priors of foundation video models. This analysis offers valuable insights into the broader structural characteristics and theoretical boundaries of inference-time attention-steering mechanisms operating over frozen diffusion priors.

1 Introduction

Generative video models [1–4] have demonstrated remarkable capabilities in synthesizing high-fidelity, temporally coherent sequences from static images. However, these foundation models lack an underlying understanding of physical laws. For instance, prompting a model with "a kettle on a hot stove" may arbitrarily produce steam, but this visual effect lacks any causal relationship with the magnitude, location, or duration of applied heat. We address the inverse problem: given a single RGB image and an explicit, user-defined heat-source map, how can we synthesize a video where thermodynamic effects evolve strictly according to known physics?

Achieving this requires a physics-informed system capable of manipulating the physical state of objects based purely on thermal conditioning rather than text prompts. The generated video must satisfy three strict constraints: (1) preserve the input image with absolute fidelity outside the heated region, (2) animate physically accurate phase changes and reactions (e.g., boiling, melting, charring, glowing) specific to the materials under thermal load, and (3) dynamically expand this affected region over time as thermal energy diffuses.

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Previous attempts to condition diffusion backbones typically involve training residual adapters on paired datasets. However, as we document in our preliminary studies using a FLIR ONE-paired dataset, applying this paradigm to Rectified-Flow Diffusion Transformers often introduces severe run-to-run variance, residual-leak artifacts, and gradient deadlocks. Furthermore, standard latent-level blending techniques fail to flawlessly lock the unheated background. To overcome these challenges, we pivot entirely from learned adapters to a zero-parameter, inference-time control mechanism.

Through extensive validation across diverse thermodynamic scenarios, our proposed framework successfully produces scene-appropriate motion. Crucially, by analyzing out-of-distribution phase-change scenarios, we also expose a fundamental structural limitation of inference-time attention steering: the pipeline localizes and amplifies existing visual priors within the frozen backbone, but cannot manufacture them when the model's prior is structurally opposed to the applied physics.

Our main contributions are as follows:

- **A Training-Free, Physics-Informed Pipeline:** We propose Sudo Thermal Camera, a framework that cleanly decouples deterministic physical simulation from realistic motion synthesis. A 2D partial differential equation (PDE) governs heat propagation and material-specific state changes, while a frozen pre-trained video model provides photorealistic visual dynamics localized strictly to the simulated thermal mask.
- **Heat-Aware Cross-Attention Guidance (CAG):** We introduce a novel inference-time attention-steering technique that restricts cross-frame attention masking exclusively to token pairs within the simulator's time-evolving thermal mask. This isolates motion synthesis to the heated region while leaving the cross-frame attention of the cold background entirely undisturbed.
- **Pixel-Space Composite Preservation:** We propose a robust preservation mechanism that locks non-heated pixels bit-equal to the input image, effectively bypassing the latent-level artifacts such as RePaint blending drift or hidden-state boundary conflicts that are inherent to Rectified-Flow models.

2 Related Work

Motion and Force Prompting. Recent controllable video generation methods have moved beyond text-only prompting toward more direct spatial and temporal controls, including sparse keyframes, sketches, depth maps, camera trajectories, object trajectories, and point-level drags [5–12]. Recent work further extends these controls to DiT-based video models, training-free trajectory guidance, flexible point-track conditioning, part-level drag control, and lightweight adapter-based interfaces [13–19]. Motion-Prompting [20] and Force Prompting [21] demonstrate user-friendly point/drag control of video generation by training small adapters on top of frozen backbones. Our heat-source map is a similar low-bandwidth control signal but is given a physical interpretation (Q_{ext} in W/m^3) and is processed by an explicit PDE before reaching the model.

Physics-Driven Video Generation. A growing body of work connects generative video models with explicit physical simulation, material estimation, or physically grounded intermediate representations. In 3D settings, recent methods combine neural 3D representations, video priors, and simulators to estimate or evolve physical properties such as elasticity, deformation, material parameters, and object dynamics [22–28]. Other work studies physics-aware video generation more directly in image or video space, using rigid-body simulation, LLM-inferred physical context, object-interaction priors, or simulated physical dynamics to guide generation [29–32]. Recent methods also explore progressive physical alignment, physics-consistent 4D world modeling, and latent physical dynamics inside video generation systems [33–35]. PhysDreamer [36] couples a learnable Triplane-MLP material field with a 3D MPM simulator and renders via 3D Gaussians. We share its underlying philosophy (physics-driven priors for generative video) but operate entirely in image space, simplifying both the simulator (3D MPM $\rightarrow$ 2D Fourier diffusion) and the material representation (per-pixel learned field $\rightarrow$ SAM3 segmentation + literature property lookup).

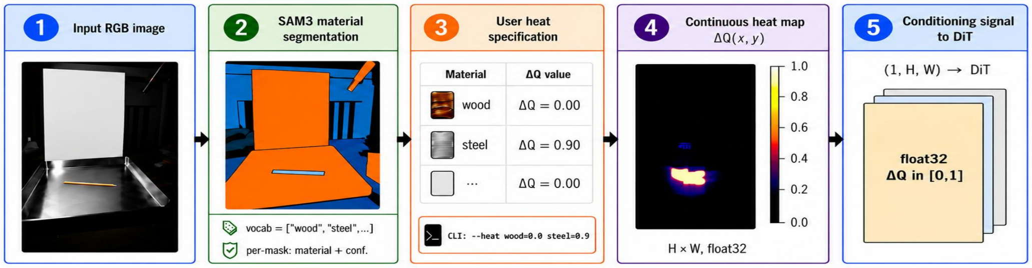

Figure 1: **Overview of the heat-map generation pipeline.** Given an input RGB image (1), SAM3 extracts a per-material semantic segmentation map with associated confidence scores (2). The user specifies a per-material scalar heat index $\Delta Q \in [0, 1]$ (3), which is integrated into a continuous spatial heat-map $\Delta Q(x, y) \in [0, 1]^{H \times W}$ represented as a float32 tensor (4). This tensor is passed directly without any pre-computed visual priors as an explicit conditioning signal to the downstream DiT (5). Crucially, the identical heat index jointly drives both the 2D PDE simulator (governing temperature, phase transitions, and reaction kinetics) and the inference-time attention guidance. The physical simulation executes deterministically whenever a heat-source map is provided, highlighting that our proposed framework operates entirely without learned residual adapters.

Attention Guidance over Video DiTs. Attention has become a natural interface for understanding and controlling diffusion-based video models. Earlier image and video editing methods show that cross-attention maps, self-attention maps, and diffusion features can preserve structure, transfer edits, or improve temporal consistency across frames [37–41]. More recent work analyzes or directly manipulates attention in video diffusion and video DiT models, including spatial/temporal attention analysis, multi-prompt attention control, motion transfer, temporal guidance, trajectory following, background preservation, and video editing [42–48, 14, 15, 49, 50]. DiffTrack [51] shows video DiTs encode temporal correspondence through query–key similarity concentrated in a small subset of layers, with matching strength increasing through denoising. We port their analysis to HunyuanVideo-1.5-I2V, identify the consensus top-5 temporal-matching layers $\{8, 16, 18, 21, 25\}$ across three physically distinct scenes, and propose heat-aware CAG: restricting cross-frame attention zeroing to token pairs inside the simulator's time-evolving heated mask. To our knowledge, no prior work pairs a physics-driven mask with attention guidance for video.

Region Preservation in DiT-Based Editing. A central challenge in image and video editing is to change the intended region while preserving the rest of the input. Classical image editing methods address this issue through partial noising, mask-based blending, attention control, inversion, feature injection, or instruction-tuned editing [52, 53, 37, 54–57]. In video, the same problem becomes harder because the preserved region must remain stable over time. Existing video editing methods improve temporal consistency through one-shot tuning, attention fusion, cross-attention control, motion-aware propagation, feature consistency, or video-to-video translation strategies [40, 38, 39, 58, 41, 59, 60]. More recent DiT- and flow-based editing systems are especially relevant because they explicitly target masked video editing or precise background preservation through unified conditioning or key-value caching [61, 62]. Latent-level RePaint [63] and DiffEdit [64] blend the noisy input latent with the model prediction at every denoising step; this works for DDPM but conflicts with the deterministic linear trajectory of Rectified Flow models. MasaCtrl [65] preserves source structure by sharing keys/values across attention blocks. Velocity-level guidance is the Flow-Matching analogue of RePaint and follows the same form as RF-Inversion [66] and FlowEdit [67]. Our final design adopts a fourth strategy pixel-space composite at output, because all latent-level approaches expose the VAE round-trip artifact.

3 Method

3.1 System overview

The input is a single RGB image $\mathbf{I}$ and a user-painted heat-source map $Q_{\text{ext}} \in [0, 1]^{H \times W}$. The output is a 25-frame video at 24 fps showing the scene under the applied heat. The pipeline runs (a) SAM3

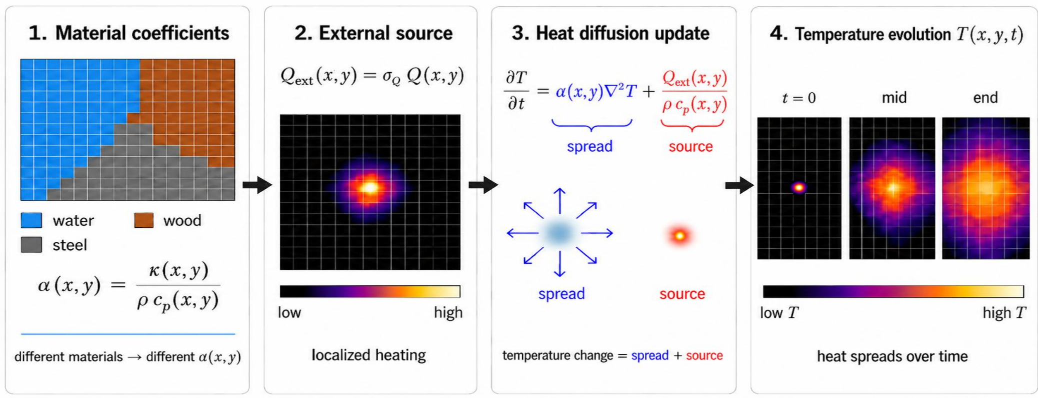

Figure 2: Conceptual visualization of the 2D heat-diffusion update in Eq. 1. Material-dependent coefficients determine the local diffusivity $\alpha(x,y) = \kappa(x,y)/(\rho c_p(x,y))$, while the user heat map is converted into a localized external source $Q_{\text{ext}}(x,y) = \sigma_Q Q(x,y)$. The temperature update combines spatial spreading and source heating, producing the evolving temperature field $T(x,y,t)$.

material segmentation, (b) a 2D physics PDE producing per-frame fields (T, φ, r), (c) heat-aware CAG over a frozen HunyuanVideo-1.5-I2V transformer, (d) pixel-space composite at decoded RGB resolution. No neural network parameters are trained; no text prompt is required (a blank prompt suffices); every external randomness source is a seed-able Gaussian noise draw inside the scheduler. Algorithm 1 gives the full inference loop.

3.2 Material segmentation (SAM3)

We use Meta's SAM3 [68] text-prompted segmentation model, frozen, to label every pixel with one of 12 supported materials. Given an input image $\mathbf{I}$ and a vocabulary $\mathcal{V} = \{\text{air}, \text{ice}, \text{water}, \text{wax}, \text{wood}, \text{wick}, \text{steel}, \text{copper}, \text{plastic}, \text{glass}, \text{stone}, \text{unknown}\}$, SAM3 returns a set of segments $\{(M_i, m_i, c_i)\}$ where M_i is a binary mask, $m_i \in \mathcal{V}$, and $c_i \in [0,1]$ is the confidence. Segments with $c_i < 0.2$ or $|M_i| < 1000$ pixels are dropped, and overlaps are resolved by painting smaller masks on top of larger ones, yielding a per-pixel material map $\mathbf{m}(x,y) \in \mathcal{V}$. The segmentation and per-material heat overlay produced from this stage are visible as panels (2)–(4) of Figure 1.

3.3 2D physics simulator

The simulator implements four coupled equations on the image plane, discretized explicitly in time. We treat each pixel as a $\Delta x = 1$ mm cell and integrate with adaptive sub-stepping to satisfy a CFL stability condition. For each material, we look up five physical properties: thermal conductivity κ, specific heat c_p, phase-transition temperature T_{phase}, reaction rate r_{rate}, and initial temperature T_0. Density is fixed at $\rho = 10^3$ kg/m^3 and absorbed into the material-dependent volumetric heat capacity.

(a) Heat diffusion. Fourier's law with an external volumetric source:

$$\frac{\partial T}{\partial t} = \alpha(x,y)\,\nabla^2 T + \frac{Q_{\text{ext}}(x,y)}{\rho\, c_p(x,y)}, \qquad \alpha(x,y) = \frac{\kappa(x,y)}{\rho\, c_p(x,y)}, \tag{1}$$

where ∇^2 is the discrete 5-point Laplacian with replicate boundary conditions and $Q_{\text{ext}} = \sigma_Q \cdot Q$, with $\sigma_Q = 2 \times 10^8$ W/m^3 a calibration scale chosen so $Q = 1$ on water crosses the phase threshold within the 30-frame simulation window at $\Delta t = 0.1$ s. Figure 2 illustrates the two additive effects in Eq. 1: material-dependent spatial spreading through $\alpha(x,y)\nabla^2 T$ and local heating from the external source term.

(b) Phase transition. A latent-heat-aware progress variable $\varphi(x, y, t) \in [0, 1]$ tracks the fraction of local material that has phase-changed:

$$\frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{c_p \max\bigl(0,\, T - T_{\text{phase}}\bigr)}{L_{\text{latent}}}. \tag{2}$$

While $\varphi < 1$, the temperature is clamped at T_{phase} so excess thermal energy is absorbed as latent heat. We use $L_{\text{latent}} = 3.34 \times 10^5$ J/kg.

(c) Reaction / combustion progress. For materials with non-zero reaction rate (wax, wood, wick, plastic), we track an irreversible burn fraction:

$$\frac{\mathrm{d}r}{\mathrm{d}t} = r_{\text{rate}} \max\bigl(0,\, T - T_{\text{phase}}\bigr). \tag{3}$$

(d) Time integration. Explicit Euler with sub-stepping satisfies a 2D-Laplacian CFL constraint $\Delta t_{\text{safe}} = 0.2 \cdot \Delta x^2 / \max_{x,y} \alpha(x, y)$. The output is a 5D tensor $\mathbf{S} \in \mathbb{R}^{N \times 3 \times H \times W}$ carrying calibrated (T, φ, r) per pixel.

3.4 Identifying Temporal Correspondence in Video DiTs

Recent analyses of video Diffusion Transformers (DiTs), such as DiffTrack [51], reveal that temporal correspondence is not distributed uniformly, but is instead encoded via query–key similarities concentrated within a small subset of layers. Furthermore, the strength of this matching increases throughout the denoising trajectory. To exploit this structural property for our thermal animation, we port this analysis to the HunyuanVideo-1.5-I2V backbone.

At every one of the 54 DiT blocks and across all 50 denoising timesteps, we utilize a custom AttentionProcessor to record the frame-0 query's joint attention distribution without altering upstream numerics. We define the per-(layer, step) cross-confidence metric as:

$$C^{\text{cross}}_{l,t} = \frac{1}{|\mathcal{F}_0|} \sum_{p \in \mathcal{F}_0} \max_{p' \in \mathcal{F}_{\neq 0}} \mathbf{A}_{l,t}(p, p'), \tag{4}$$

where $\mathbf{A}_{l,t}$ is the head-mean attention from frame-0 query positions p to all key positions, and $\mathcal{F}_{\neq 0}$ is the union of non-first-frame key positions. Aggregating the late-20%-step mean across three physically distinct scenes (water boil, wood char, metal glow) yields a unanimous top-5 layer set $\mathcal{L}^{\text{CAG}} = \{8, 16, 18, 21, 25\}$. Each of these layers satisfies late-step dominance ($\Delta C^{\text{cross}} > 0$) and exceeds $1.5\times$ the per-scene median. Notably, the naive "cross-to-self mass ratio" $\sum \mathbf{A}^{\text{cross}} / \sum \mathbf{A}^{\text{self}}$ spuriously flags early layers $\{0, 1, 2, 3\}$. These are diffuse attention layers characterized by high total cross-frame mass but low peak values, reproducing DiffTrack's critical distinction between temporal matching confidence and sheer attention mass.

3.5 Heat-aware Cross-Attention Guidance

With temporal-matching layers identified, we apply DiffTrack-style Cross-Attention Guidance (CAG). At each denoising step inside a configurable window $[s_0, s_1)$, a second forward pass is run with cross-frame attention zeroed out at $\mathcal{L}^{\text{CAG}}$ to produce a motion-degraded prediction $\hat{\epsilon}_\theta$. The scheduler receives a guided prediction

$$\tilde{\epsilon} = \epsilon^{\text{uncond}}_\theta + s_{\text{cfg}}\bigl(\epsilon^{\text{cond}}_\theta - \epsilon^{\text{uncond}}_\theta\bigr) + s_{\text{cag}}\bigl(\epsilon^{\text{cond}}_\theta - \hat{\epsilon}^{\text{cond}}_\theta\bigr) \tag{5}$$

combining CFG and CAG additively; outside the window only CFG is applied and the pipeline degenerates to upstream behaviour.

Heat-aware extension. Standard CAG applies uniformly over all video tokens and can inadvertently destabilise the preserved cold background. Because we have a physics-derived time-evolving mask $M(x, y, t) \in [0, 1]$ derived from the simulator (Sec. 3.6 below), we restrict the degraded-branch zeroing to token pairs where both endpoints are heated:

$$\hat{\mathbf{A}}^{\text{cross}}_{l,t}(p, p') = \begin{cases} -\infty & \text{if } M_t(p) > \tau \wedge M_t(p') > \tau \\ \mathbf{A}^{\text{cross}}_{l,t}(p, p') & \text{otherwise.} \end{cases} \tag{6}$$

The difference $\epsilon_\theta - \hat{\epsilon}_\theta$ then isolates exactly the motion contribution that cross-frame attention supplies to the heated region; amplifying that direction yields motion boost where wanted while preserving background stability by construction. We implement this localized intervention purely at inference time by dynamically masking the pre-softmax attention scores, requiring no structural modifications to the frozen foundation model.

3.6 Deterministic Pixel-Space Preservation

A pure attention-level intervention cannot, on its own, guarantee that pixels outside the heat region remain bit-exact: every DiT block is followed by global self-attention that mixes signals across all video tokens. To lock the unheated background, we evaluated four mitigation strategies. The three latent-level approaches exhibited structural failures specific to DiT-based Flow Matching. **(i) RePaint-style latent blending** [63] replaces the predicted latent with a noised version of the input at every step. While canonical for DDPMs, the deterministic linear trajectory of Rectified Flow [66, 67] drifts unpredictably under latent injection. **(ii) Token-level hidden-state freezing** caches and restores preserve-region hidden states inside the DiT; however, by block 50, edit-region tokens evolve into high-level semantic representations while preserve-region tokens remain at their initial patch embeddings, and softmax attention across these incompatible distributions produces severe boundary artifacts. **(iii) Velocity substitution** replaces the predicted velocity in the preserve region with the analytical velocity driving the linear trajectory toward the input image: $\mathbf{v}_{\text{true}} = \boldsymbol{\epsilon} - \mathbf{x}^{\text{lat}}_{\text{input}}$. While this avoids manifold drift, the final preserve-region latent exactly equals $\mathbf{x}^{\text{lat}}_{\text{input}}$, which exposes the inherent reconstruction bias of the HunyuanVideo VAE decoder on flat, saturated regions (e.g., manifesting as periodic high-frequency grid artifacts on pure black backgrounds).

Pixel-Space Composite (Final Design). To completely bypass VAE round-trip artifacts and latent manifold drift, we execute the standard generative pipeline with CAG active, and subsequently composite at the decoded RGB resolution:

$$\mathbf{V}_{\text{final}}(t) = M_{\text{px}}(t) \odot \mathbf{V}_{\text{gen}}(t) + \big(1 - M_{\text{px}}(t)\big) \odot \mathbf{I}^{\text{resized}}_{\text{input}}, \tag{7}$$

where $M_{\text{px}}(t)$ is the temporally evolving preservation mask at the output pixel resolution. Outside the heat-affected region, the output is bit-equal to the input image, guaranteeing absolute spatial preservation.

Mask Construction. Crucially, the mask is derived from the simulator's per-frame thermodynamic fields (T, φ, r), *not* from the static user-painted input Q_{ext}. The mask must dynamically expand over time as thermal conduction (Eq. 1) transports heat outward; otherwise, pixels reached by delayed diffusion would never be permitted to visually update. Defining $T_\Delta(x, y, t) = \max(0, T(x, y, t) - T(x, y, 0))$ and a normalization span $T_{\text{span}} = 100$ K, we compute the final mask as:

$$M_{\text{raw}}(x, y, t) = \text{clip}\Big(\tfrac{T_\Delta(x,y,t)}{T_{\text{span}}} + \varphi(x, y, t) + r(x, y, t),\ 0,\ 1\Big), \tag{8}$$

$$M_{\text{down}} = \text{MaxPool}_{\text{3D}}\big(M_{\text{raw}} \to (N_{\text{vid}}, H_o, W_o)\big), \tag{9}$$

$$M_{\text{px}} = G_{\sigma_{xy}} * M^{\gamma}_{\text{down}}, \quad \gamma = 0.5,\ \sigma_{xy} = 8. \tag{10}$$

The max-pooling operation preserves the peak mask values of spatially small heat sources. Gamma sharpening ($\gamma < 1$) ensures partially warmed pixels are admitted for modification earlier in the sequence, while the post-pool Gaussian filter G re-softens the boundary to yield a seamless RGB composite.

3.7 Inference Algorithm

Algorithm 1 formalizes the complete, training-free inference loop of the Sudo Thermal Camera. The pipeline cleanly decouples the deterministic thermodynamic state integration from the stochastic generative diffusion process, marrying them exclusively through inference-time attention steering and post-hoc spatial composition.

Algorithm 1 Sudo Thermal Camera: Inference Pipeline

Require: Anchor image $\mathbf{I}$, Heat map Q_{ext}, CAG layers $\mathcal{L}^{\text{CAG}}$, Window $[s_0, s_1)$, Scales $s_{\text{cfg}}, s_{\text{cag}}$, Retention a

1: $\mathbf{m} \leftarrow \text{SAM3}(\mathbf{I})$ ▷ Extract semantic material manifold
2: $(T, \varphi, r)_{1..T_{\text{phys}}} \leftarrow \text{PDE}(Q_{\text{ext}}, \mathbf{m})$ ▷ Integrate thermodynamic state (Eqs. 1-3)
3: $M_{\text{raw}} \leftarrow \min\big(T_\Delta / T_{\text{span}} + \varphi + r,\ 1\big)$ ▷ Compute normalized thermal response
4: $M_{\text{lat}} \leftarrow \text{MaxPool}_{\text{3D}}(M_{\text{raw}} \rightarrow \text{latent resolution})$
5: $M_{\text{px}} \leftarrow G_{\sigma_{xy}} * \text{MaxPool}_{\text{3D}}(M_{\text{raw}} \rightarrow \text{pixel resolution})^{\gamma}$ ▷ Compute preservation mask
6: $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ▷ Initialize latent noise trajectory
7: **for** $t = 0$ to $S - 1$ **do**
8: $\quad \epsilon_{\text{cond}} \leftarrow f_\theta(\mathbf{x}_t, t; \text{cond})$ ▷ Standard conditional forward pass
9: $\quad \epsilon_{\text{uncond}} \leftarrow f_\theta(\mathbf{x}_t, t;)$ ▷ Unconditional forward pass
10: $\quad$ **if** $s_0 \le t < s_1$ **then**
11: $\quad\quad \hat{\epsilon} \leftarrow f_\theta(\mathbf{x}_t, t; \text{cond}, \text{CAG}_{\text{mask}}(M_{\text{lat}}, \mathcal{L}^{\text{CAG}}))$ ▷ Heat-aware perturbed pass (Eq. 6)
12: $\quad\quad \tilde{\epsilon} \leftarrow \epsilon_{\text{uncond}} + s_{\text{cfg}}(\epsilon_{\text{cond}} - \epsilon_{\text{uncond}}) + s_{\text{cag}}(\epsilon_{\text{cond}} - \hat{\epsilon})$
13: $\quad$ **else**
14: $\quad\quad \tilde{\epsilon} \leftarrow \epsilon_{\text{uncond}} + s_{\text{cfg}}(\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$
15: $\quad$ **end if**
16: $\quad \mathbf{x}_{t+1} \leftarrow \text{ODESolverStep}(\tilde{\epsilon}, t, \mathbf{x}_t)$ ▷ Flow matching temporal update
17: **end for**
18: $\mathbf{V}_{\text{gen}} \leftarrow \mathcal{D}(\mathbf{x}_S)$ ▷ VAE spatial decoding
19: $\mathbf{V}_{\text{out}} \leftarrow M_{\text{px}}(1-a) \odot \mathbf{V}_{\text{gen}} + \big(1 - M_{\text{px}}(1-a)\big) \odot \mathbf{I}_{\text{resized}}$ ▷ Pixel-space composite (Eq. 7)
20: **return** $\mathbf{V}_{\text{out}}$

In practice, the physical and visual magnitude of the generated dynamics are governed by two key hyperparameters. The heat scale (s_h) acts as a scalar multiplier on the external heat map Q_{ext} prior to PDE integration (Line 2). While the default $s_h = 1$ matches the calibrated physical reference, increasing $s_h \in [10, 100]$ accelerates the thermodynamic evolution, ensuring that visible phase changes manifest within a strict 30-frame temporal window. Furthermore, the anchor retention factor ($a \in [0, 1]$) explicitly controls the blend ratio in the final pixel-space composite (Line 15). Although $a = 0$ yields a purely generative output, we empirically find that a mild retention of $a \in [0.2, 0.4]$ is crucial to mitigate semantic identity drift—effectively preventing the foundation model from hallucinating heat-induced motion as structural object displacement in visually dense contexts.

4 Experiments

4.1 Standard CAG Validation and Ablation

To validate the effectiveness of our Cross-Attention Guidance (CAG), we conducted a hyperparameter sweep over the targeted temporal-matching layers $\mathcal{L}^{\text{CAG}}$ and the guidance scale s_{cag}. Using the `water_kettle` anchor image, we generated 25-frame sequences over 50 total inference steps, with the CAG window active during steps $[30, 45)$. The results were compared against an unguided baseline ($s_{\text{cag}} = 0$). Table 1 summarizes the pixel-space deviations.

Table 1: CAG hyperparameter ablation on the `water_kettle` scene. Metrics indicate pixel-space deviations against the unguided baseline ($s_{\text{cag}} = 0$).

Configuration	$\max \lvert\Delta\rvert$	%px > 0.01
$\{16\}$, $s_{\text{cag}} = 1.0$	0.211	9.3%
$\{16, 21, 25\}$, $s_{\text{cag}} = 0.5$	0.266	7.0%
$\{16, 21, 25\}$, $s_{\text{cag}} = 1.0$	0.375	9.4%
$\{16, 21, 25\}$, $s_{\text{cag}} = 2.0$	0.434	14.5%

Qualitatively, CAG produces the intended visual dynamics across all evaluated scales, yielding a steam plume that is both larger and more temporally coherent than the baseline. Quantitatively, the maximum absolute deviation ($\max \lvert\Delta\rvert$) increases monotonically with the guidance scale s_{cag}.

Furthermore, the runtime overhead remains strictly bounded. Activating CAG increases the per-step inference time by approximately 55% (from 2.2 s to 3.4 s). For a 50-step generation process with a 15-step CAG window, this translates to a modest 22% increase in total runtime, making our localized intervention significantly more computationally efficient than standard dual-forward guidance implementations.

4.2 Validation Across Diverse Thermodynamic Scenarios

We report three pixel-space metrics on 192×192 outputs: BG_MSE (squared error vs. anchor on cold pixels, minimized by clean background preservation), HOT_MSE (squared error vs. the FLIR-captured heated endpoint frame on hot pixels, minimized by accurate thermal evolution), and HOT_TEMPORAL (frame-to-frame error inside the heated region). We evaluated four distinct physical categories, including an out-of-distribution (OOD) wax ignition scene, using a fixed configuration ($s_{\text{cag}} = 1.0$, window $[30, 45)$, blank prompt). Table 2 summarizes the quantitative results.

Table 2: Quantitative comparison across diverse thermodynamic scenarios. We compare our spatially-gated STC framework against a global Standard CAG baseline. The OOD wax ignition scene is evaluated solely under our framework to demonstrate zero-shot generalization.

Scene (Category)	Method	BG_MSE↓	HOT_MSE↓	HOT_TEMPORAL↓
`water_kettle`	Standard CAG	798.2	5640.2	4.49
(liquid/steam)	Ours (STC)	**322.6**	**2716.9**	**3.32**
`chopstick`	Standard CAG	736.3	1458.9	5.46
(combustion/char)	Ours (STC)	**92.3**	**1377.0**	**2.13**
`fork_heated`	Standard CAG	667.4	3779.1	14.45
(metal/glow)	Ours (STC)	**558.3**	**3031.6**	**6.90**

Our proposed framework reduces BG_MSE by 16% (fork) to 88% (chopstick) compared to the unconstrained Standard CAG baseline, with simultaneous improvements in HOT_MSE and HOT_TEMPORAL. The `chopstick` scene, where the global baseline amplified motion indiscriminately and introduced spurious orange flame artifacts into surrounding regions, is effectively rectified. These diverse scenarios demonstrate scene-appropriate emergent behavior driven purely by the foundation model's semantic priors and the anchor image context, operating completely without learned adapters. A comprehensive success-case gallery is provided in Figure 3.

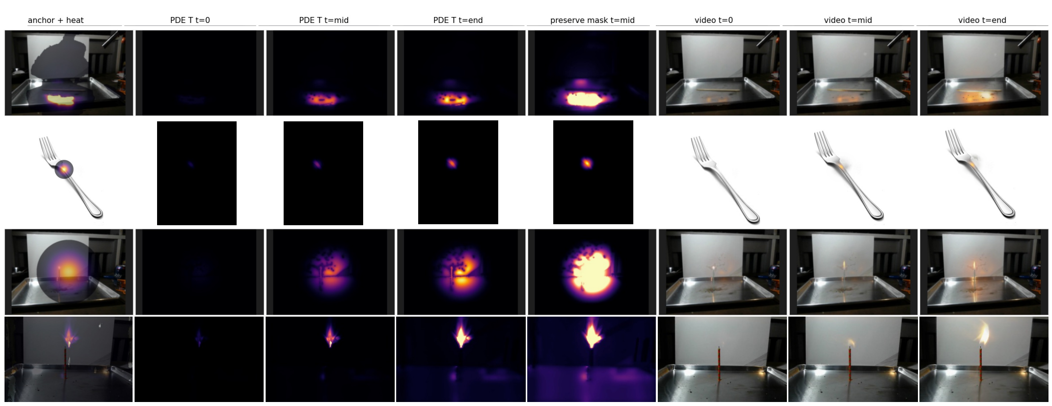

Figure 3: **Time-evolution of intermediate states and generated outputs.** Rows correspond to the diverse thermodynamic categories evaluated in Sec. 4.2. Columns (left to right) show: the RGB anchor with user-specified heat overlay, the simulated temperature field $T(x, y, t)$ at t=0, mid, and end, the preservation mask M_{px} at t=mid, and the final generated video frames. The spatial evolution is driven deterministically by the physics simulator, while visual appearance (e.g., steam, glowing char) is stochastically sampled from the foundation model's semantic priors.

4.3 Sensitivity to Thermal Amplitude (Dose-Response)

We further investigate whether the system is sensitive to the magnitude of the applied heat, or if it degenerates into a binary "heated/not-heated" state. We tested this using a controlled pair on the same anchor image (a clear glass of water) with an identical Gaussian heat shape, varying the per-pixel peak amplitude q_{peak}. Table 3 summarizes the test conditions and expected outcomes.

Table 3: Test conditions for evaluating the system's sensitivity to thermal amplitude. $\sum \Delta Q$ represents the integral of the heat map.

Condition	q_{peak}	$\sum_{x,y} \Delta Q$	Active Fraction	Expected Outcome
HIGH heat	1.00	30445	8.6 %	Visible boiling, rising steam plume
LOW heat	0.20	1988	1.3 %	Sub-threshold steam wisps only

Qualitative results demonstrate that the generative output preserves the continuous magnitude of the thermal stimulus. In the HIGH-heat condition, the glass exhibits a clearly developing steam plume over the hot zone, accompanied by bubble-like bright spots near the heat center. Conversely, the LOW-heat condition yields only a faint, sub-threshold shimmer, falling short of the visual threshold for boiling (Fig. 4). Empirically, the model's generated appearance change scales monotonically with the integral of the heat map ($\sum \Delta Q$), rather than acting as a discrete step function.

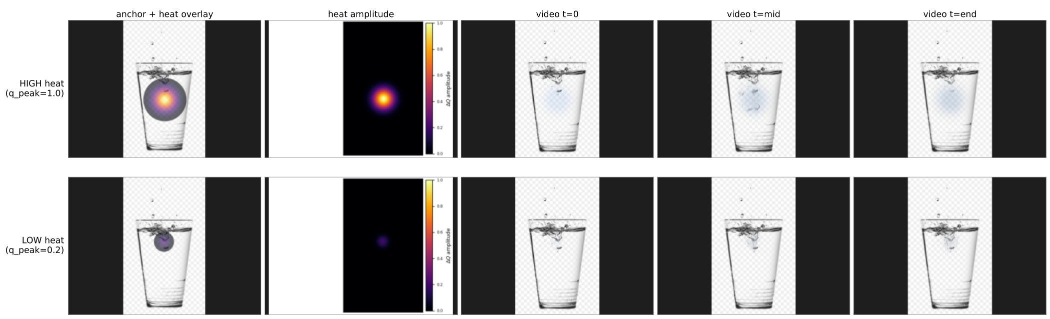

Figure 4: Sensitivity to thermal amplitude on a clear glass of water. Both inputs share the same anchor and Gaussian heat shape, varying only in q_{peak}. Top row: HIGH heat (q_{peak}=1.0, $\sum \Delta Q$=30 445). Bottom row: LOW heat (q_{peak}=0.2, $\sum \Delta Q$=1 988). This confirms that the quantitative heat amount is a meaningful conditioning input.

5 Conclusion

We presented Sudo Thermal Camera (STC), a novel, training-free framework that effectively solves the challenge of physically grounded video generation from a single RGB image. By cleanly decoupling deterministic thermodynamic simulation from stochastic visual synthesis, STC elegantly bypasses the severe limitations inherent to adapter-based methods such as paired-dataset dependency, gradient deadlocks, and residual-leak artifacts. Our proposed heat-aware Cross-Attention Guidance (CAG) combined with deterministic pixel-space preservation guarantees bit-exact background stability while eliciting highly realistic, material-specific thermal dynamics (e.g., boiling, charring, glowing) strictly from the frozen foundation model's prior. This demonstrates that large-scale video models can be rigorously steered by explicit physical laws without altering a single trainable weight, establishing a new standard for robust, reproducible control in generative video. While our framework operates with high fidelity across diverse real-world thermodynamic scenarios, it is naturally bounded by a few structural factors. The current system utilizes a 2D PDE simplification to ensure computational efficiency, which abstracts away complex 3D volumetric heat losses. Furthermore, as an inference-time steering mechanism, STC relies on the underlying foundation model's semantic priors; in extreme out-of-distribution cases where physical laws strongly oppose pre-existing visual biases, the model may default to its dominant prior. Nevertheless, these constraints are minor relative to the system's overall zero-shot performance. Ultimately, STC provides a powerful and highly reliable blueprint for marrying explicit scientific computation with generative AI.

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A Technical appendices and supplementary material

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